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Problem Set 6:

Comprehensive Solutions



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This problem set covers Chapters 26, 27, and 28 of Peng Ding's *A First Course in Causal Inference*, together with additional problems on bounds, principal stratification, and mediation analysis.

1.1 Exercises from Chapter 26: Principal Stratification

1.1.1 Background Results

Let Z be a binary treatment, M a binary post-treatment variable, Y the outcome, and X pretreatment covariates. Define principal strata by

$$S = (M(1), M(0)).$$

The principal stratification average causal effect in stratum (m_1, m_0) is

$$\tau(m_1, m_0) = \mathbb{E}\{Y(1) - Y(0) \mid M(1) = m_1, M(0) = m_0\}.$$

Assumption 1.1 (CRE with an intermediate variable)

$$Z \perp \{M(1), M(0), Y(1), Y(0), X\}.$$

Assumption 1.2 (strong monotonicity)

$$M(0) = 0.$$

Assumption 1.3 (principal ignorability under strong monotonicity)

$$\mathbb{E}\{Y(0) \mid M(1) = 1, X\} = \mathbb{E}\{Y(0) \mid M(1) = 0, X\}.$$

Under Assumption 1.2, define

$$\pi(X) = \mathbb{P}\{M(1) = 1 \mid X\}, \quad \pi = \mathbb{P}\{M(1) = 1\}.$$

Theorem 1.1 (principal score identification) *Under Assumptions 1.1–1.3,*

(a) the conditional and marginal principal scores,

$$\pi(X) = \mathbb{P}\{M(1) = 1 \mid X\}, \quad \pi = \mathbb{P}\{M(1) = 1\},$$

can be identified by

$$\pi(X) = \mathbb{P}(M = 1 \mid Z = 1, X), \quad \pi = \mathbb{P}(M = 1 \mid Z = 1).$$

(b) provided the denominators are positive, the principal stratification average causal effects are identified by

$$\tau(1, 0) = \mathbb{E}(Y \mid Z = 1, M = 1) - \frac{\mathbb{E}\{\pi(X)Y \mid Z = 0\}}{\pi}$$

and

$$\tau(0, 0) = \mathbb{E}(Y \mid Z = 1, M = 0) - \frac{\mathbb{E}\{(1 - \pi(X))Y \mid Z = 0\}}{1 - \pi}.$$

26.1 Complete the proof for $\tau(0, 0)$

Prove the result for $\tau(0, 0)$ in theorem 1.1.

Solution to Problem 26.1: Under strong monotonicity, $M(0) = 0$, so the observed cell $(Z = 1, M = 0)$ is exactly the principal stratum with $M(1) = 0$ and $M(0) = 0$. By consistency and randomization,

$$\mathbb{E}(Y \mid Z = 1, M = 0) = \mathbb{E}\{Y(1) \mid M(1) = 0\}.$$

It remains to identify $\mathbb{E}\{Y(0) \mid M(1) = 0\}$. Principal ignorability implies

$$\mathbb{E}\{Y(0) \mid M(1) = 0, X\} = \mathbb{E}\{Y(0) \mid X\},$$

and hence

$$\mathbb{E}\{(1 - M(1))Y(0) \mid X\} = (1 - \pi(X))\mathbb{E}\{Y(0) \mid X\}.$$

By randomization, consistency, and the tower property,

$$\begin{aligned} \frac{\mathbb{E}\{(1 - \pi(X))Y \mid Z = 0\}}{1 - \pi} &= \frac{\mathbb{E}\{(1 - \pi(X))Y(0)\}}{1 - \pi} \\ &= \frac{\mathbb{E}[(1 - \pi(X))\mathbb{E}\{Y(0) \mid X\}]}{1 - \pi} \\ &= \frac{\mathbb{E}\{(1 - M(1))Y(0)\}}{\mathbb{P}\{M(1) = 0\}} \\ &= \mathbb{E}\{Y(0) \mid M(1) = 0\}. \end{aligned}$$

Subtracting the two identified terms gives the claimed formula for $\tau(0, 0)$. $\square$

26.2 Principal score method under monotonicity

Replace strong monotonicity with ordinary monotonicity $M(1) \geq M(0)$. Let

$$\begin{aligned}\pi_{(m_1, m_0)}(X) &= \mathbb{P}\{M(1) = m_1, M(0) = m_0 \mid X\}, \\ \pi_{(m_1, m_0)} &= \mathbb{E}\{\pi_{(m_1, m_0)}(X)\}.\end{aligned}$$

Assume principal ignorability:

$$\begin{aligned}\mathbb{E}\{Y(1) \mid S = (1, 0), X\} &= \mathbb{E}\{Y(1) \mid S = (1, 1), X\}, \\ \mathbb{E}\{Y(0) \mid S = (1, 0), X\} &= \mathbb{E}\{Y(0) \mid S = (0, 0), X\}.\end{aligned}$$

Derive the principal score identification formulas.

Theorem 1.2 (identification under monotonicity) *Under complete randomization, monotonicity $M(1) \geq M(0)$, and the principal ignorability conditions displayed above, the conditional principal scores are identified by*

$$\begin{aligned}\pi_{(0,0)}(X) &= \mathbb{P}(M = 0 \mid Z = 1, X), \\ \pi_{(1,1)}(X) &= \mathbb{P}(M = 1 \mid Z = 0, X), \\ \pi_{(1,0)}(X) &= \mathbb{P}(M = 1 \mid Z = 1, X) - \mathbb{P}(M = 1 \mid Z = 0, X),\end{aligned}$$

and the marginal principal scores are identified by

$$\begin{aligned}\pi_{(0,0)} &= \mathbb{P}(M = 0 \mid Z = 1), \\ \pi_{(1,1)} &= \mathbb{P}(M = 1 \mid Z = 0), \\ \pi_{(1,0)} &= \mathbb{P}(M = 1 \mid Z = 1) - \mathbb{P}(M = 1 \mid Z = 0).\end{aligned}$$

Moreover,

$$\tau(1, 0) = \mathbb{E}\{w_{1,(1,0)}(X)Y \mid Z = 1, M = 1\} - \mathbb{E}\{w_{0,(1,0)}(X)Y \mid Z = 0, M = 0\},$$

$$\tau(0, 0) = \mathbb{E}(Y \mid Z = 1, M = 0) - \mathbb{E}\{w_{0,(0,0)}(X)Y \mid Z = 0, M = 0\},$$

and

$$\tau(1, 1) = \mathbb{E}\{w_{1,(1,1)}(X)Y \mid Z = 1, M = 1\} - \mathbb{E}(Y \mid Z = 0, M = 1),$$

where

$$\begin{aligned}w_{1,(1,0)}(X) &= \frac{\pi_{(1,0)}(X)}{\pi_{(1,0)}(X) + \pi_{(1,1)}(X)} \bigg/ \frac{\pi_{(1,0)}}{\pi_{(1,0)} + \pi_{(1,1)}}, \\ w_{0,(1,0)}(X) &= \frac{\pi_{(1,0)}(X)}{\pi_{(1,0)}(X) + \pi_{(0,0)}(X)} \bigg/ \frac{\pi_{(1,0)}}{\pi_{(1,0)} + \pi_{(0,0)}}, \\ w_{0,(0,0)}(X) &= \frac{\pi_{(0,0)}(X)}{\pi_{(1,0)}(X) + \pi_{(0,0)}(X)} \bigg/ \frac{\pi_{(0,0)}}{\pi_{(1,0)} + \pi_{(0,0)}}, \\ w_{1,(1,1)}(X) &= \frac{\pi_{(1,1)}(X)}{\pi_{(1,0)}(X) + \pi_{(1,1)}(X)} \bigg/ \frac{\pi_{(1,1)}}{\pi_{(1,0)} + \pi_{(1,1)}}.\end{aligned}$$

Solution to Problem 26.2: Under $M(1) \geq M(0)$, the possible strata are $(1, 1), (1, 0), (0, 0)$. The principal score part of theorem 1.2 follows from consistency and randomization:

$$\begin{aligned}\pi_{(0,0)}(X) &= \mathbb{P}(M = 0 \mid Z = 1, X), \\ \pi_{(1,1)}(X) &= \mathbb{P}(M = 1 \mid Z = 0, X), \\ \pi_{(1,0)}(X) &= \mathbb{P}(M = 1 \mid Z = 1, X) - \mathbb{P}(M = 1 \mid Z = 0, X).\end{aligned}$$

The marginal versions are

$$\begin{aligned}\pi_{(0,0)} &= \mathbb{P}(M = 0 \mid Z = 1), \\ \pi_{(1,1)} &= \mathbb{P}(M = 1 \mid Z = 0), \\ \pi_{(1,0)} &= \mathbb{P}(M = 1 \mid Z = 1) - \mathbb{P}(M = 1 \mid Z = 0).\end{aligned}$$

The useful posterior weights are

$$\begin{aligned}w_{1,(1,0)}(X) &= \frac{\pi_{(1,0)}(X)}{\pi_{(1,0)}(X) + \pi_{(1,1)}(X)} \bigg/ \frac{\pi_{(1,0)}}{\pi_{(1,0)} + \pi_{(1,1)}}, \\ w_{0,(1,0)}(X) &= \frac{\pi_{(1,0)}(X)}{\pi_{(1,0)}(X) + \pi_{(0,0)}(X)} \bigg/ \frac{\pi_{(1,0)}}{\pi_{(1,0)} + \pi_{(0,0)}}, \\ w_{0,(0,0)}(X) &= \frac{\pi_{(0,0)}(X)}{\pi_{(1,0)}(X) + \pi_{(0,0)}(X)} \bigg/ \frac{\pi_{(0,0)}}{\pi_{(1,0)} + \pi_{(0,0)}}, \\ w_{1,(1,1)}(X) &= \frac{\pi_{(1,1)}(X)}{\pi_{(1,0)}(X) + \pi_{(1,1)}(X)} \bigg/ \frac{\pi_{(1,1)}}{\pi_{(1,0)} + \pi_{(1,1)}}.\end{aligned}$$

For example, among units with $(Z, M) = (1, 1)$, the latent strata are $(1, 0)$ and $(1, 1)$. The first factor in $w_{1,(1,0)}(X)$ is the posterior probability of stratum $(1, 0)$ given $M(1) = 1$ and X , and the second factor normalizes the weight to have mean one in the observed mixture. Hence, using principal ignorability for $Y(1)$,

$$\mathbb{E}\{w_{1,(1,0)}(X)Y \mid Z = 1, M = 1\} = \mathbb{E}\{Y(1) \mid S = (1, 0)\}.$$

Likewise, among $(Z, M) = (0, 0)$, the latent strata are $(1, 0)$ and $(0, 0)$, and principal ignorability for $Y(0)$ gives

$$\mathbb{E}\{w_{0,(1,0)}(X)Y \mid Z = 0, M = 0\} = \mathbb{E}\{Y(0) \mid S = (1, 0)\}.$$

Subtracting these two identified terms yields

$$\tau(1, 0) = \mathbb{E}\{w_{1,(1,0)}(X)Y \mid Z = 1, M = 1\} - \mathbb{E}\{w_{0,(1,0)}(X)Y \mid Z = 0, M = 0\}.$$

The same argument applied to the strata $(0, 0)$ and $(1, 1)$ gives

$$\tau(0, 0) = \mathbb{E}(Y \mid Z = 1, M = 0) - \mathbb{E}\{w_{0,(0,0)}(X)Y \mid Z = 0, M = 0\},$$

and

$$\tau(1, 1) = \mathbb{E}\{w_{1,(1,1)}(X)Y \mid Z = 1, M = 1\} - \mathbb{E}(Y \mid Z = 0, M = 1).$$

□

26.3 Principal score method in observational studies

Replace complete randomization by

$$Z \perp \{M(1), M(0), Y(1), Y(0)\} \mid X,$$

and let $e(X) = \mathbb{P}(Z = 1 \mid X)$. Under strong monotonicity and assumption 1.3, prove the observational-study identification result.

Theorem 1.3 (identification in observational studies) *Assume*

$$Z \perp \{M(1), M(0), Y(1), Y(0)\} \mid X,$$

positivity $0 < e(X) < 1$, strong monotonicity, assumption 1.3, and positive principal strata probabilities. Then

$$\pi(X) = \mathbb{P}(M = 1 \mid Z = 1, X), \quad \pi = \mathbb{E}\{\mathbb{P}(M = 1 \mid Z = 1, X)\},$$

and the principal stratification average causal effects are identified by

$$\tau(1, 0) = \mathbb{E}\left\{\frac{M}{\pi} \frac{Z}{e(X)} Y\right\} - \mathbb{E}\left\{\frac{\pi(X)}{\pi} \frac{1 - Z}{1 - e(X)} Y\right\},$$

and

$$\tau(0, 0) = \mathbb{E}\left\{\frac{1 - M}{1 - \pi} \frac{Z}{e(X)} Y\right\} - \mathbb{E}\left\{\frac{1 - \pi(X)}{1 - \pi} \frac{1 - Z}{1 - e(X)} Y\right\}.$$

Solution to Problem 26.3: Strong monotonicity implies $\pi(X) = \mathbb{P}\{M(1) = 1 \mid X\} = \mathbb{P}(M = 1 \mid Z = 1, X)$, and the marginal probability is

$$\pi = \mathbb{E}\{\mathbb{P}(M = 1 \mid Z = 1, X)\}.$$

For the first term of $\tau(1, 0)$,

$$\begin{aligned} \mathbb{E}\left\{\frac{M}{\pi} \frac{Z}{e(X)} Y\right\} &= \frac{1}{\pi} \mathbb{E}\left[\mathbb{E}\left\{\frac{Z}{e(X)} M(1) Y(1) \mid X, M(1), Y(1)\right\}\right] \\ &= \frac{\mathbb{E}\{M(1) Y(1)\}}{\pi} = \mathbb{E}\{Y(1) \mid M(1) = 1\}. \end{aligned}$$

For the second term, by ignorability and principal ignorability,

$$\begin{aligned} \mathbb{E}\left\{\frac{\pi(X)}{\pi} \frac{1 - Z}{1 - e(X)} Y\right\} &= \frac{\mathbb{E}\{\pi(X) Y(0)\}}{\pi} \\ &= \frac{\mathbb{E}\{M(1) Y(0)\}}{\pi} = \mathbb{E}\{Y(0) \mid M(1) = 1\}. \end{aligned}$$

Their difference identifies $\tau(1, 0)$.

Similarly,

$$\mathbb{E}\left\{\frac{1 - M}{1 - \pi} \frac{Z}{e(X)} Y\right\} = \mathbb{E}\{Y(1) \mid M(1) = 0\},$$

and

$$\mathbb{E} \left\{ \frac{1 - \pi(X)}{1 - \pi} \frac{1 - Z}{1 - e(X)} Y \right\} = \mathbb{E}\{Y(0) \mid M(1) = 0\}.$$

Subtracting identifies

$$\tau(0, 0) = \mathbb{E} \left\{ \frac{1 - M}{1 - \pi} \frac{Z}{e(X)} Y \right\} - \mathbb{E} \left\{ \frac{1 - \pi(X)}{1 - \pi} \frac{1 - Z}{1 - e(X)} Y \right\}.$$

□

1.2 Exercises from Chapter 27: Mediation Analysis

1.2.1 Definitions and Assumptions

Let $Y(z, m)$ denote the potential outcome if the treatment is set to z and the mediator to m , and let $M(z)$ denote the potential mediator. The natural direct and indirect effects are

$$\text{NDE} = \mathbb{E}\{Y(1, M(0)) - Y(0, M(0))\}, \quad \text{NIE} = \mathbb{E}\{Y(1, M(1)) - Y(1, M(0))\}.$$

Assumption 1.4 (no treatment-outcome confounding)

$$Z \perp Y(z, m) \mid X$$

for all z, m .

Assumption 1.5 (no mediator-outcome confounding)

$$M \perp Y(z, m) \mid (Z, X)$$

for all z, m .

27.2 Sequential randomization and joint randomization

Show that Assumptions 1.4 and 1.5 are equivalent to

$$(Z, M) \perp Y(z, m) \mid X$$

for all z, m .

Solution to Problem 27.2: Assume first that Assumptions 1.4 and 1.5 hold. For any fixed (z, m) , write $Y^* = Y(z, m)$. The two assumptions give

$$f(y^* \mid Z, M, X) = f(y^* \mid Z, X) = f(y^* \mid X),$$

where the first equality uses no mediator-outcome confounding and the second uses no treatment-outcome confounding. Thus $(Z, M) \perp Y^* \mid X$.

Conversely, if $(Z, M) \perp Y^* \mid X$, then

$$f(y^* \mid Z, M, X) = f(y^* \mid X).$$

Averaging over $M \mid Z, X$ gives $f(y^* \mid Z, X) = f(y^* \mid X)$, so $Z \perp Y^* \mid X$. Combining this with the displayed equality gives $f(y^* \mid Z, M, X) = f(y^* \mid Z, X)$, so $M \perp Y^* \mid (Z, X)$. Hence the two formulations are equivalent, up to the usual positivity conditions. $\square$

27.8 Treatment-outcome interaction in the outcome model

Suppose

$$\begin{cases} \mathbb{E}(M \mid Z, X) = \beta_0 + \beta_1 Z + \beta_2^T X, \\ \mathbb{E}(Y \mid Z, M, X) = \theta_0 + \theta_1 Z + \theta_2 M + \theta_3 ZM + \theta_4^T X. \end{cases}$$

Show that

$$\text{NDE} = \theta_1 + \theta_3\{\beta_0 + \beta_2^T \mathbb{E}(X)\}, \quad \text{NIE} = (\theta_2 + \theta_3)\beta_1,$$

and describe estimation with IID data.

Solution to Problem 27.8: Let $\mu_Y(z, m, X) = \mathbb{E}(Y \mid Z = z, M = m, X)$.

By the mediation formula, under the standard mediation assumptions,

$$\text{NDE} = \mathbb{E} \left[\int \{\mu_Y(1, m, X) - \mu_Y(0, m, X)\} dF_{M \mid Z=0, X}(m) \right],$$

and

$$\text{NIE} = \mathbb{E} \left[\int \mu_Y(1, m, X) \{dF_{M \mid Z=1, X}(m) - dF_{M \mid Z=0, X}(m)\} \right].$$

For a fixed X and mediator value m , the outcome-model contrast is

$$\{\theta_0 + \theta_1 + \theta_2 m + \theta_3 m + \theta_4^T X\} - \{\theta_0 + \theta_2 m + \theta_4^T X\} = \theta_1 + \theta_3 m.$$

Averaging over $M(0)$ and X gives

$$\text{NDE} = \theta_1 + \theta_3 \mathbb{E}\{M(0)\} = \theta_1 + \theta_3\{\beta_0 + \beta_2^T \mathbb{E}(X)\}.$$

For the natural indirect effect, hold $Z = 1$ in the outcome model:

$$\mathbb{E}(Y \mid Z = 1, M = m, X) = \theta_0 + \theta_1 + (\theta_2 + \theta_3)m + \theta_4^T X.$$

Therefore

$$\text{NIE} = (\theta_2 + \theta_3) \mathbb{E}\{M(1) - M(0)\} = (\theta_2 + \theta_3)\beta_1.$$

With IID data, fit the mediator model and the outcome model by least squares, estimate $\mathbb{E}(X)$ by the sample mean, and plug in the estimated coefficients. A nonparametric bootstrap over units gives standard errors and confidence intervals. $\square$

1.3 Exercises from Chapter 28: Controlled Direct Effects

28.2 Observational studies with a multi-valued treatment

Let $Z \in \{1, \dots, K\}$ be a multi-valued treatment, $Y(k)$ the potential outcome under level k , and assume

$$Z \perp \{Y(1), \dots, Y(K)\} \mid X, \quad e_k(X) = \mathbb{P}(Z = k \mid X) > 0.$$

Define $\mu_k = \mathbb{E}\{Y(k)\}$ and $\mu_k(X) = \mathbb{E}(Y \mid Z = k, X)$. Prove the outcome regression, IPW, and doubly robust formulas.

Solution to Problem 28.2: By ignorability and consistency,

$$\mu_k(X) = \mathbb{E}(Y \mid Z = k, X) = \mathbb{E}\{Y(k) \mid X\}.$$

Therefore

$$\mathbb{E}\{\mu_k(X)\} = \mathbb{E}[\mathbb{E}\{Y(k) \mid X\}] = \mu_k.$$

For the IPW formula, conditioning on X and the potential outcomes,

$$\mathbb{E}\left\{\frac{\mathbb{I}(Z = k)Y}{e_k(X)} \mid X, Y(1), \dots, Y(K)\right\} = \frac{e_k(X)Y(k)}{e_k(X)} = Y(k),$$

so

$$\mathbb{E}\left\{\frac{\mathbb{I}(Z = k)Y}{e_k(X)}\right\} = \mu_k.$$

For working models $e_k(X, \alpha)$ and $\mu_k(X, \beta)$, define

$$\mu_k^{\text{dr}} = \mathbb{E}\{\mu_k(X, \beta)\} + \mathbb{E}\left[\frac{\mathbb{I}(Z = k)\{Y - \mu_k(X, \beta)\}}{e_k(X, \alpha)}\right].$$

If $\mu_k(X, \beta) = \mu_k(X)$, then the augmentation has conditional mean zero given X , regardless of the treatment model, so $\mu_k^{\text{dr}} = \mu_k$. If $e_k(X, \alpha) = e_k(X)$, then

$$\begin{aligned} \mu_k^{\text{dr}} &= \mathbb{E}\{\mu_k(X, \beta)\} + \mathbb{E}[\mathbb{E}\{Y(k) - \mu_k(X, \beta) \mid X\}] \\ &= \mathbb{E}\{\mu_k(X, \beta)\} + \mathbb{E}\{\mu_k(X) - \mu_k(X, \beta)\} = \mu_k. \end{aligned}$$

Thus the formula is doubly robust. Any contrast $\tau_C = \sum_{k=1}^K C_k \mu_k$ with $\sum_k C_k = 0$ is then identified by applying the same estimator to each μ_k . $\square$

1.4 Additional Problems

2. Bounds on CACE under one-sided noncompliance

Consider an encouragement experiment with one-sided noncompliance, $D_i(0) = 0$. Let $C = \{D(1) = 1\}$ denote the complier stratum, and sup-

pose the outcome is binary. Derive bounds on the complier effect without the exclusion restriction.

Solution to Problem 2: With one-sided noncompliance, units with $(Z, D) = (1, 1)$ are exactly compliers. Let

$$\pi_c = \mathbb{P}(C) = \mathbb{P}(D = 1 \mid Z = 1), \quad \mu_{1c} = \mathbb{E}(Y \mid Z = 1, D = 1).$$

Without the exclusion restriction, the control potential mean among compliers,

$$\mu_{0c} = \mathbb{E}\{Y(0, 0) \mid C\},$$

is not identified because the observed control group mixes compliers and never-takers:

$$\mathbb{E}(Y \mid Z = 0) = \pi_c \mu_{0c} + (1 - \pi_c) \mu_{0n},$$

where $\mu_{0n} = \mathbb{E}\{Y(0, 0) \mid D(1) = 0\}$. Since Y is binary, $0 \leq \mu_{0n} \leq 1$, and therefore

$$\max \left\{ 0, \frac{\mathbb{E}(Y \mid Z = 0) - (1 - \pi_c)}{\pi_c} \right\} \leq \mu_{0c} \leq \min \left\{ 1, \frac{\mathbb{E}(Y \mid Z = 0)}{\pi_c} \right\}.$$

Thus the complier effect

$$\tau_c = \mathbb{E}\{Y(1, 1) - Y(0, 0) \mid C\} = \mu_{1c} - \mu_{0c}$$

has bounds

$$\mu_{1c} - \min \left\{ 1, \frac{\mathbb{E}(Y \mid Z = 0)}{\pi_c} \right\} \leq \tau_c \leq \mu_{1c} - \max \left\{ 0, \frac{\mathbb{E}(Y \mid Z = 0) - (1 - \pi_c)}{\pi_c} \right\}.$$

With the exclusion restriction, the never-taker mean under control can be learned from treated non-takers, and these bounds collapse to the usual Wald identification formula. $\square$

3. Principal stratification and individual causal effect

Let Z_i be binary treatment and Y_i binary outcome. Define

$$\text{ACE} = \mathbb{E}\{Y_i(1) - Y_i(0)\}, \quad \tau = \mathbb{P}\{Y_i(1) > Y_i(0)\}.$$

Solution to Problem 3:

1. In the drug-survival example, ACE measures the average change in survival probability, while τ measures the proportion of individuals for whom the drug is beneficial at the individual level. As written, the requested example with $\tau < 0$ is impossible because $\tau = \mathbb{P}\{Y(1) > Y(0)\}$ is a probability. Thus no numerical example can satisfy the prompt literally. If the intended contrast is that

ACE = 0 can coexist with positive individual benefit, a meaningful example is

$$\mathbb{P}\{Y(1), Y(0) = (1, 0)\} = 0.25, \quad \mathbb{P}\{Y(1), Y(0) = (0, 1)\} = 0.25,$$

and the remaining 0.50 split arbitrarily between (1, 1) and (0, 0). Then

$$\text{ACE} = 0.25 - 0.25 = 0, \quad \tau = 0.25.$$

The average effect is zero because helped and harmed units cancel, but a quarter of the population benefits.

2. Let

$$\pi_{y_1 y_0} = \mathbb{P}\{Y(1) = y_1, Y(0) = y_0\}.$$

Since the possible principal strata are (1, 1), (1, 0), (0, 1), (0, 0),

$$\text{ACE} = \mathbb{E}\{Y(1) - Y(0)\} = \pi_{10} - \pi_{01},$$

and

$$\tau = \mathbb{P}\{Y(1) > Y(0)\} = \mathbb{P}\{Y(1) = 1, Y(0) = 0\} = \pi_{10}.$$

3. First consider randomized trials. Let

$$p_1 = \mathbb{P}(Y = 1 \mid Z = 1) = \mathbb{P}\{Y(1) = 1\},$$

$$p_0 = \mathbb{P}(Y = 1 \mid Z = 0) = \mathbb{P}\{Y(0) = 1\}.$$

Writing $t = \pi_{10}$, the four principal-stratum probabilities satisfy

$$\pi_{11} = p_1 - t, \quad \pi_{00} = 1 - p_0 - t, \quad \pi_{01} = t - (p_1 - p_0).$$

The constraints $\pi_{y_1 y_0} \geq 0$ imply

$$t \leq p_1, \quad t \leq 1 - p_0, \quad t \geq p_1 - p_0, \quad t \geq 0.$$

Therefore the sharp randomization bounds are

$$\max\{0, p_1 - p_0\} \leq \tau \leq \min\{p_1, 1 - p_0\}.$$

Without randomization, write

$$\begin{aligned} \pi_{10} &= \mathbb{P}\{Z = 1, Y(1) = 1, Y(0) = 0\} \\ &\quad + \mathbb{P}\{Z = 0, Y(1) = 1, Y(0) = 0\}. \end{aligned}$$

The first term is bounded by $\mathbb{P}(Z = 1, Y = 1)$, since among $Z = 1$ units the observed outcome is $Y(1)$. The second term is bounded by $\mathbb{P}(Z = 0, Y = 0)$, since among $Z = 0$ units the observed outcome is $Y(0)$. Hence

$$0 \leq \pi_{10} \leq \mathbb{P}(Z = 1, Y = 1) + \mathbb{P}(Z = 0, Y = 0).$$

□

4. Principal stratification and direct and indirect effects

Let Z be binary treatment, M binary post-treatment variable, and Y the outcome. The individual direct effects are

$$\text{CDE}_i(m) = Y_i(1, m) - Y_i(0, m), \quad \text{NDE}_i(z) = Y_i(1, M_i(z)) - Y_i(0, M_i(z)).$$

Solution to Problem 4:

1. If $\text{NDE}_i(0) = 0$ for all i , then

$$Y_i(1, M_i(0)) - Y_i(0, M_i(0)) = 0.$$

In a principal stratum with $M_i(1) = M_i(0) = m$, we have $M_i(0) = m$ and, by composition, $Y_i(1) = Y_i(1, m)$ and $Y_i(0) = Y_i(0, m)$. Therefore

$$Y_i(1) - Y_i(0) = Y_i(1, m) - Y_i(0, m) = \text{NDE}_i(0) = 0$$

for every unit in the stratum. Averaging gives

$$\mathbb{E}\{Y_i(1) - Y_i(0) \mid M_i(1) = M_i(0) = m\} = 0$$

for $m = 0, 1$.

2. If $\text{CDE}_i(m) = 0$ for all i and $m = 0, 1$, then for units with $M_i(1) = M_i(0) = m$,

$$Y_i(1) - Y_i(0) = Y_i(1, m) - Y_i(0, m) = \text{CDE}_i(m) = 0.$$

Again the principal causal effect in the unchanged mediator strata is zero.

□

5. Analysis of anxiety in media framing effects

Brader, Valentino, and Suhay conducted a randomized media-framing experiment. The treatment is $Z = 1$ when the story combines the cost frame with the Hispanic immigrant picture. The mediator is anxiety, and the outcome is the binary indicator for requesting that an anti-immigration message be sent to Congress.

Solution to Problem 5:

1. The natural direct and indirect effects are

$$\begin{aligned} \text{NDE} &= \mathbb{E}\{Y(1, M(0)) - Y(0, M(0))\}, \\ \text{NIE} &= \mathbb{E}\{Y(1, M(1)) - Y(1, M(0))\}. \end{aligned}$$

For identification, I use consistency, positivity, no treatment-outcome confounding, no mediator-outcome confounding, no treatment-mediator confounding, and the cross-world independence

$$Y(z, m) \perp M(z') \mid X.$$

For a continuous mediator, the mediation formula gives

$$\text{NDE} = \mathbb{E}_X \int \{\mu_Y(1, m, X) - \mu_Y(0, m, X)\} f_{M|Z=0, X}(m) dm,$$

$$\text{NIE} = \mathbb{E}_X \int \mu_Y(1, m, X) \{f_{M|Z=1, X}(m) - f_{M|Z=0, X}(m)\} dm.$$

2. I used the **framing** data from the CRAN **mediation** package. The variable **anx** is coded as an ordered factor; I recoded it as 1 = not anxious at all, 2 = a little anxious, 3 = somewhat anxious, and 4 = very anxious. The fitted models were

$$M \mid Z, X \sim N(\beta_0 + \beta_1 Z + \beta_2^T X, \sigma_M^2),$$

$$\text{logit}\{\mathbb{P}(Y = 1 \mid Z, M, X)\} = \theta_0 + \theta_1 Z + \theta_2 M + \theta_4^T X,$$

with $X = (\text{age}, \text{educ}, \text{gender}, \text{income})$. I evaluated the mediation integrals by plug-in prediction and used 1000 nonparametric bootstrap resamples for variances and percentile confidence intervals. The script is **analysis_framing.R**.

Effect	Estimate	Variance	SE	2.5%	97.5%
NDE	0.0346	0.00367	0.0606	-0.0837	0.1591
NIE	0.0674	0.00053	0.0229	0.0254	0.1184
Total effect	0.1021	0.00441	0.0664	-0.0293	0.2278

The mediator model estimated a negative effect of treatment on anxiety ($\hat{\beta}_1 = -0.447$), and the outcome model estimated a negative association between anxiety and requesting the congressional message ($\hat{\theta}_2 = -0.780$ on the logit scale). Their product intuition explains why the estimated indirect effect is positive.

3. The estimated natural indirect effect is about 0.067, with a bootstrap 95% interval $[0.025, 0.118]$, suggesting evidence that the treatment changes the outcome through the anxiety channel. In this coding, the treatment lowers reported anxiety, and lower anxiety is associated with a higher probability of requesting the message. The natural direct effect is smaller and imprecise: the 95% interval includes zero. The total estimated effect is positive but also imprecise, so most of the detectable signal in this specification comes from the mediated path.

□